

Next-Gen AI Compiler

Agents On The Control Plane

Agents Own Search · Orchestration · Synthesis.
Compilers Own Lowering · Legality · Measure · Fallback.

Ask: Is The Hybrid Prediction (~2027-31) The Right Bet For Your Roadmap?

Expert Briefing · ~30-35 min + Q&A · Prediction-first

Agenda

- 
1. Trends · Keep Substrate
 2. Verdict + Architecture
 3. Roadmap Checkpoints
 4. Commercial Blockers + Gaps
 5. Technical Prediction (Techniques)
 6. Stance · Ask · Appendix

Trend backdrop — two stacks converging



Compilers for AI
mature · fragmented substrate

AI for compilers
rapid hybrid growth

Next-gen $\neq$ throw away LLVM/MLIR — make them **agent-addressable**.

Six active trends

A — Hybrid guidance

Free IR rewrite fragile
(mlirAgent / UC Berkeley)
Constrain: hints / metadata / passes
AgentCompile (CityU)
HintPilot (ZJU)
Meta LLM Compiler

B — RL → agents

CompilerGym → tool agents
Heuristics as shippable C++
Workflow compile · freeze · place
Compiler-R1 (ISCAS)
Magellan (Google DeepMind)
FlowCompile (UMass/MIT)

C — MLIR + Triton

PyTorch → Inductor → Triton
StableHLO → XLA/IREE
Peak: vendor libs / Tile / CUTLASS
Meta · OpenXLA/Google
OpenAI Triton · NVIDIA

D — Kernel agents

KernelBench (Stanford/Princeton)
One-shot often <20%; fusion hard
GEAK (AMD)
KernelLLM (Meta)
AgentCompile (CityU)

E — Verify in the loop

Unit/golden → numerical
→ Alive2-class local formal
Weak on GPU races / FP noise
Alive2 (UIUC lineage)
LLM-VeriOpt · stacked admit

F — Broader object

Mid-decode diagnosis
FMware: prompts / agents / knobs
Agents as compiler engineers
Compiler.next (Queen's)
Magellan (Google)
Claude C (Anthropic)

Keep substrate, change control plane

Preserve

deterministic lowering
library kernels
build-CI regressability
local formal checks

Adopt

advisory search
multi-agent orchestration
heuristic synthesis
profile / verifier feedback

**Anti-pattern: replace opt/Inductor
with unconstrained LLM codegen**

Executive verdict

Agents reshape the **control plane**
more than they replace the **data plane**.

Control plane: search / orchestrate / synthesize

Data plane: lower / legality / measure / fallback

Compilers for AI
× AI for compilers

Hybrid LLM-compiler loops

4th job Tier A:
bring-up / codesign

Will not happen soon

- Unconstrained LLM replaces Inductor/opt
- Autonomous chip tape-out via compiler agents (C10)

Four agent jobs — architecture spine

(a) Online

propose →
measure
admit

CompileIQ / GEAK
ACCLAIM / HintPilot

(b) Offline

evolve heuristics
→ C++ / MLGO

Magellan
AlphaEvolve

(c) Eng.

oracle-gated
pull request
/ change

Archer

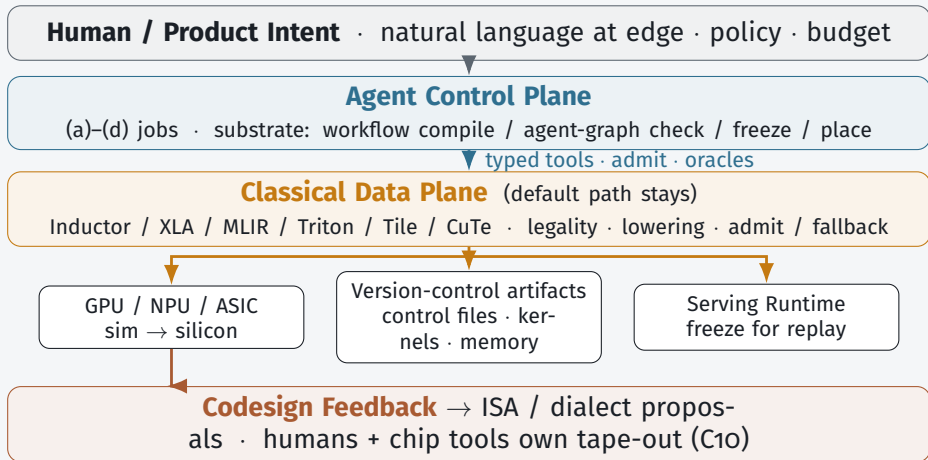
(d) Bring-up

coverage →
performance
sim + silicon

TritorX
KernelEvolve

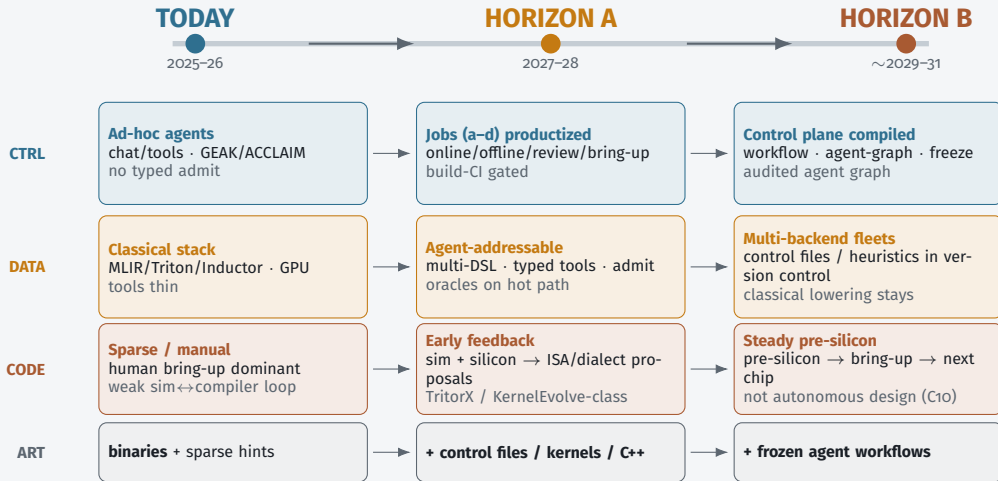
Control plane jobs sit *above* classical lowering — not instead of it.

Architecture — Target Stack



Invariant: LLM guides search — does not silently define unchecked executable behavior.

Architecture evolution — component changes



Components thicken left→right; data plane stays; agent graph becomes compiled & amortized.

Data plane — ~6–7 abstraction bands (not one)

Must-have today (L1–L6)

L1	Framework	capture · dynamism
L2	Portable graph	StableHLO-class · shard and
L3	Mid-IR	MLIR · layout · passes
L4	Kernel DSL	Triton/Helion/Tile/CuTe
L5	Backend-ISA	PTX · bring-up
L6	Runtime-serve	CUDA Graphs · KV

Lo* CPU/LLVM optional · agents sit *above* bands via tools

L7* Fleet / cluster

maturing: place + collectives
today often L2–L3 + runtime

Not a new IR band

power/energy → objective + oracle
\$/token · SLOs → control policy
safety → provenance / admit

Lean (A6/S6)

keep bands; agents unify *contracts*,
not one mega-IR

One universal cost model? — No (Horizon A)

Why not one model

Different legality / oracles per band
Choosing the level *is* the product
Cost models stay *local* (MLGO, Ansor)
Future passes need new measured labels

If bands do not consolidate

Agent compile schema (admit hash)
Typed tools / MCP-class servers
Dialect + oracle + objective plugins
Placement / fleet plugins (L7)

Size: local advisors often KB–MB · cross-band *proposer* $\approx$ 7B–70B+ IR LLM (still only a prior)

Bottleneck = labeled (program, action, HW, energy, SLO) tuples — not parameter count.

Bet: small per-band costs + optional large orchestrator — not one mega-cost-model for L1–L7.

E2E-optimal-seeking — reshape around F

Product fitness F (latency · energy · \$/token · quality · cluster) — Pareto/constrained, not $\sum$ local costs

E2E search controller — joint/bilevel policy · credit across bands · budget/stop/freeze

L2–L3

L4–L5

L6

L7*

bands = legality / lower surfaces · local cost = proposal prior only

Physical F -admit: serving A/B → pinned traces
→ energy/fleet · freeze or classical fallback

Claimed

e2e-optimal-seeking control plane
joint search + F -admit + freeze

Not claimed

unique closed-form global optimum
complexity bound $\neq$ stop trying

Merge $\neq$ replace — three levels

M1 Soft merge (Horizon A $\rightarrow$ B)

e2e controller + admit/trace over band tools · data plane stays · agents do *not* replace compiler

M2 Search-surface merge (contested / partial $\sim$ 2029–31)

fewer agent training IRs (C4) · lowers still multi-band · still hybrid

M3 Hard replace — *not* claimed thru Horizon B ($\sim$ 2031)

default path with *no* classical admit/fallback · only if C6-A settles

Perception trap: “I only talk to an agent” is often M1+freeze (classical lower still ran in CI).
Lean (A8 / C6-B): merge the *optimizer* under *F*; do not delete the execution substrate.

What ships / does not by 2028

Predicted to ship

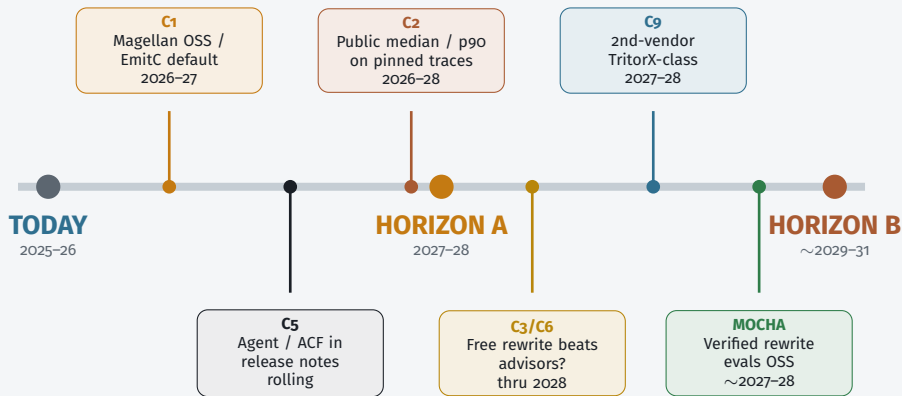
- Agent-addressable tool interfaces
- Hot-path specialize (not silent default-all)
- Magellan *and* MLGO both live
- Oracle pull-request review in serious orgs
- Coverage-first ASIC bring-up
- Triton-family primary; multi-DSL rising

Does not ship

- Unconstrained LLM replaces opt/Inductor
- One agent IR for all vendors
- Kernel agents uniformly beat eager on fusion ladders
- Autonomous microarch tape-out

Horizon A success = build-CI gated specialize + oracles + freeze artifacts

Roadmap Checkpoints — when the prediction changes



Falsifiable milestones — watch settlement signals; do not average disagreements.

Checkpoint C1 — heuristics vs neural advisors

A — Magellan path

evolve shippable C++ heuristics

Evolve shippable C++
(EVOLVE-BLOCK)

Offline coding agent *is* the
control-plane output

Product = evolutionary coding agent +
review

B — MLGO / EmitC path

MLGO = learned in-tree LLVM advisors

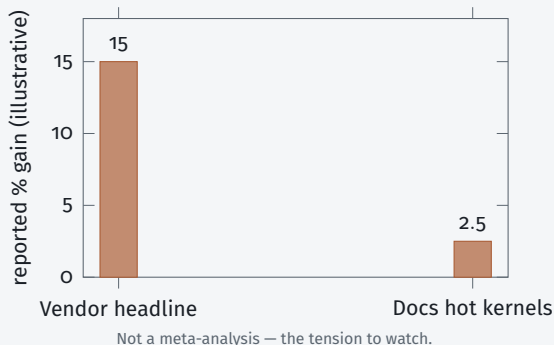
NN advisors stay in-tree

June 2026 PoR: inliner → An-
droid/Fuchsia → Chrome

Agents train/feature NNs

Settlement: public Magellan llvm patches that dis-
place MLGO or EmitC-MLGO becomes customer default
Watch LLVM syncs + OpenEvolve recipes quarterly through 2027

Checkpoints C2 + C5 — gains & default path



C2 “Agents become default” needs **median (p50) build-CI wins**, not best-kernel blogs.

p50 / p90 = 50th / 90th percentile of the gain distribution across builds.

C5 Online specialize (control files / hints) vs offline eng (Magellan/MLGO) — both may live; *which is the default flag?*

Settlement: pinned public traces + release notes listing agent / control-file workflows.

Checkpoints C₃ / C₆ / C₉ / C₁₀ — interface width & scope

C₃ — free rewrite vs advisory

Flip: shared suite where free rewrite consistently wins

⇒ wide rewrite API vs narrow ACFs / hints (current lean)

C₆ — replace vs control plane

Flip: production *default* path with no classical admit / fallback

⇒ agents replaced the compiler (survey rejects; hold hybrid)

C₉ — coverage vs peak

Flip: TritorX → KernelEvolve ladder becomes a public playbook

⇒ SKU: coverage SLA first, then performance SLA

C₁₀ — codesign vs auto tape-out

Flip: microarch primarily agent-proposed *and* compiler-oracle validated

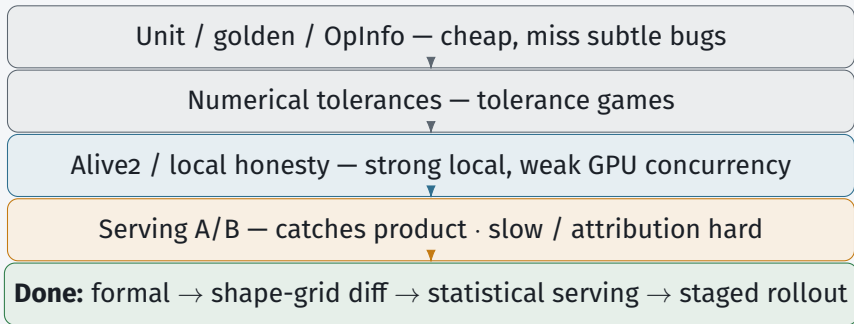
⇒ enters chip-design tooling (today: agents stress kernels/IR only)

Commercial blockers — top 5

- 1 Oracles strong enough for money
Fast-but-wrong · liability
- 2 Cost / replay / when agents run
Nondeterministic \$N compiles
- 3 Agent $\leftrightarrow$ compiler contract
Natural-language-only = demo
- 4 Distributional production evidence
No default-path A/B
- 5 Ownership · supply chain · human review
Unowned agent code in trusted base

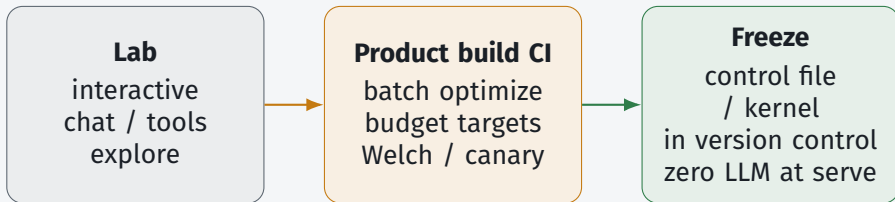
Each blocker gets one slide — productization gaps, not research wishlist.

Blocker 1 — Oracles for money



Room question: who owns false negatives when admit passes and production miscompiles?

Blocker 2 — Cost, replay, when may the agent run?



Cache key: (IR hash, hardware, compiler ver, agent policy) · budget \$/%gain

Falsifies commercially: “chat with compiler on every build” without spend/latency targets.

Blocker 3 — Agent $\leftrightarrow$ compiler contract

A. Natural language + pasted logs — demo only

B. Structured admit traces — build CI / bill of materials

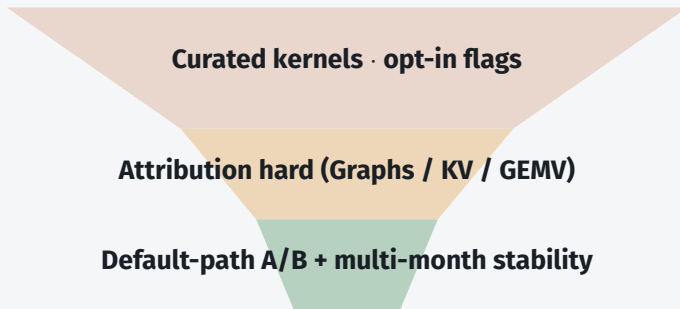
C. Typed tool interfaces — required action space

D. Hybrid — natural-language view over traces

```
admit_record = {graph_hash, hw_id, compiler_ver, action[],  
oracle[], artifact_digest, policy_id}
```

Lean: **C** + **B**; Natural language is a view, not
source of truth. ACF = portable compiler knobs.

Blocker 4 — Distributional production evidence



Marketing bar: median + 90th-percentile gains + cost-to-compile
· persistent serving throughput, not single-kernel headlines

Blocker 5 — Ownership, security, human review

Trusted-base risk

agent kernels /
heuristics enter
trusted base

Ownership

CODEOWNERS
signed provenance
sandbox

Human-review capacity

agents $\uparrow$ drafts
reviewers =
bottleneck

Lean: human CODEOWNER + signed admit + sand-
box · oracle auto-merge only for **narrow** action classes
Generic forge AI $\neq$ compiler-oracle review (C7)

Gap map — what blocks the prediction, not a wishlist

HIGH

4.1 prod evidence

4.2 correctness

4.3 cost/replay

4.4 cross-stack

4.10 benchmarks

MED-HIGH

4.5 hardware-native interfaces

4.7 training data

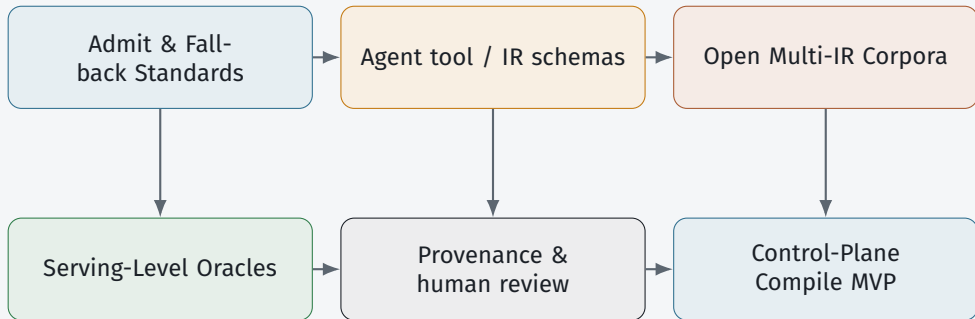
MEDIUM

4.6 workflow compile

4.8 human-review process

4.9 security

Cross-Cutting Research Agenda



Near-term research themes → next: technique map T1-T10 (exists / missing / unlocks).

Technical Prediction — Accelerate The Roadmap

To settle checkpoints and ship Horizon A,
enhance techniques *inside and outside* the compiler (T₁–T₁₀).

Within (T₁–T₅)

- T1 Typed agent ↔ compiler interfaces
- T2 Admit / fallback machinery
- T3 Control files, hints, fingerprints + replay
- T4 Heuristic hooks & in-tree advisors
- T5 Dialect / ISA feedback sinks

Outside (T₆–T₁₀)

- T6 Serving-level oracles & production A/B
- T7 Open multi-IR corpora (+ negatives)
- T8 Unified benchmark ladder
- T9 Provenance, ownership, HITL process
- T10 Agent-workflow compile / freeze / place

Enhancing only opt/Inductor/Triton internals is not enough. Next: exists · missing today · checkpoint unlocks.

Technical Prediction — Within The Compiler (T_I–T₅)

Exists · Missing · Unlocks

T₁ Typed interfaces

→ C₃, C₅, C₆

Exists: CompileIQ · ACCLAIM · mlir-opt-repl · FlashInfer Trace

Missing: portable schemas across MLIR · Triton · Tile · StableHLO

T₂ Admit / fallback

→ C₆ hybrid

Exists: AgentCompile · Archer · TritonRL verifiers · FlashInfer-Bench

Missing: shared admit *product* + trusted deterministic fallback

T₃ Control files + replay

→ C₂, C₅

Exists: CompileIQ ACFs · FlashInfer Trace + `apply()`

Missing: content-addressed keys; golden replay on model upgrade

T₄ Heuristic hooks / advisors

→ C₁

Exists: Magellan / AlphaEvolve · MLGO · EmitC PoR

Missing: settled Magellan vs MLGO *default* on named apps

T₅ Dialect / ISA sinks

→ C₉, C₁₀

Exists: TritonX · KernelEvolve · Ascend diagnosis

Missing: first-class change-*proposal* surfaces (not auto tape-out)

Technical Prediction — Outside The Compiler (T6–T10)

Exists · Missing · **Unlocks**

T6 Serving oracles / A/B

→ **C2**

Exists: unit/golden/Alive2 · **FlashInfer-Bench**+apply() · VibeServe

Missing: whole-program / GPU-race / FP; multi-month default-path A/B

T7 Multi-IR corpora

→ **Selectors**

Exists: Meta LLM Compiler · **KernelBook**→TritonRL · DR Triton

Missing: versioned MLIR / Tile / StableHLO + failed / miscompile negatives

T8 Benchmark ladder

→ **C2, C9**

Exists: KernelBench(-X) · FlashInfer-Bench serving-kernel rung

Missing: full IR→kernel→fused→serving + cost-to-compile

T9 Provenance / HITL

→ **C7**

Exists: Magellan reviewable C++ · Archer oracle review

Missing: CODEOWNERS + signed admit records + sandbox for proposals

T10 Workflow compile / freeze

→ **Horiz. B**

Exists: FlowCompile · Auto · AgentFlow · VibeServe (early)

Missing: shared agent-graph IR + fail-closed CI → Horizon B

Technical Prediction — What Each Checkpoint Needs

C1 $\leftarrow$ **T4**

Magellan-class synthesis or EmitC-MLGO customer default on named apps

C2 $\leftarrow$ **T3 + T6 + T8**

Replayable artifacts + serving oracles + ladder
FlashInfer-Bench = pressure, not settlement

C5 $\leftarrow$ **T1 + T3**

Typed interfaces + freeze-before-serve named in release notes

C3 / C6 $\leftarrow$ **T1 + T2**

Constrained tools + admit/fallback (lean: hybrid advisory)

C9 $\leftarrow$ **T5 + T8**

Coverage $\rightarrow$ perf agents + dialect sinks; need 2nd-vendor TritonX-class path

C10 $\leftarrow$ **T5**

ISA/dialect *proposals only*; humans + chip-design tools own tape-out

Technical Prediction — Critical Missing Parts Now

Highest-leverage bets to accelerate Horizon A

1. Money-grade oracle stack (T2+T6)

w/o → demos

local formal → shape-grid diff → serving statistics → staged rollout

2. Replayable artifact contract (T3)

w/o → CI rejects

control files / kernels / heuristics with cache keys + golden replay

3. Portable agent compile interface (T1)

w/o → re-glue

summaries · actions · admit records across MLIR / Triton / Tile / StableHLO

4. Open ladder + multi-IR data (T7+T8)

w/o → incomparable

correctness × speed × \$/compile + negative (failed) examples

Survey lean: T1–T5 ship as product surfaces; T6–T10
equally first-class — mostly outside classical lowering.

Org Adoption Questions

1. Online (CompileIQ) vs Offline (Magellan) — which matches your release model?

2. Oracle stack: Alive2 / golden / serving A/B — who owns false negatives?

3. Agent contract surface: LLVM / MLIR / Triton / StableHLO / Tile?

4. How do you cache/regress traces across compiler *and* model upgrades?

5. Max \$/build for a median X% win? Named maintainer per admitted artifact?

Working stance until conflicts settle

- 1 Hybrid: agents search; compilers lower (not replace)
- 2 Magellan || MLGO — parallel bets (C1)
- 3 Discount single-number speedups (C2)
- 4 Constrained actions + strong oracles (C3)
- 5 Demote generic SCM AI (C7)
- 6 Coverage→performance; no autonomous chip design (C9/C10)
- 7 ~6–7 data-plane bands; plugins if not consolidated (A6/S6)
- 8 E2E-optimal-seeking: joint search under F (A7/S7)
- 9 Soft merge M1 by Horizon B; hard replace M3 not claimed (A8/C6)

Commercial checklist — handout

Architecture

typed tools + admit
traces
version control >>
dense memory >>
scratchpad
state-machine plan for
service targets
freeze before default-
on

Trust

layered oracles
CODEOWNERS + prove-
nance
joint version pins
A/B before “prod de-
fault”

Business

sell regressable arti-
facts
+ build-CI quota
customer owns out-
puts
coverage then perfor-
mance service target
token budget per
%gain

Discussion

Thank you — hybrid bet stays the ask

1. Which checkpoint flips your investment first — C1, C2, or C9?
2. Harder commercial bar: **oracle strength** or **replay/freeze**?
3. Building job (a), (b), or (d) first — what do you refuse online?

Ask again: is the hybrid bet right for your roadmap? Watch settlement signals — do not average disagreements.

Appendix

Evidence Store Snapshot

Publications digests · commercial product signals · open repositories.
Tiered for the agentic-compiler prediction — not a full catalog.

Appendix — evidence map

Publications

~115 digests
one file per source
papers · blogs · talks
forge pages · minutes
★ = prediction-critical

Products

Commercial SKUs as
prediction signals
Tier A shapes the future
Tier B = data-plane de-
faults
not a full SKU list

Repositories

GitHub / Gerrit / forge
artifacts tiered A/B/C
A = agents reshape
compile
C = generic forge AI only
not an exhaustive forge
map

Use ★ digests + Tier A products/repos when a checkpoint or blocker is contested.

Appendix — Tier A commercial signals

Company offerings that shape what ships — agent loop / kernel surface / cloud agent

Google / DeepMind	AlphaEvolve Cloud (GA); Magellan + MLGO (heuristics <i>and</i> advisors)
NVIDIA	CompileIQ (+ agent-skills); CUDA Tile / Tile IR; TensorRT-LLM agent skills
AMD	GEAK (multi-agent Triton / Instinct kernel optimization)
Meta	LLM Compiler / KernelLLM; TritonX + KernelEvolve; Helion
FlashInfer	FlashInfer-Bench (serving-trace ladder + <code>apply()</code> into SGLang/vLLM)

Tier B baselines: TensorRT-LLM · Inductor · XLA/StableHLO · FlashInfer · Modular MAX · OpenVINO · Neuro

Appendix — Tier A open repositories

Offline / review / heuristics

OpenEvolve · HeuriGym · Archer · Compiler-R1 · HintPilot · mlirAgent

Online / kernels / bring-up

ACCLAIM · CompileIQ · GEAK · KernelAgent · KernelBench(-X)
FlashInfer-Bench · AutoKernel · Helion · TritonX / KernelEvolve

Tiers: A = agents + domain oracles change heuristics/kernels/knobs/review
B = data-plane hosts · C = generic forge AI (demoted for prediction)

Appendix — prediction-critical digests (★)

Highest-signal sources — mechanisms, not headlines

Vision / funded

Compiler 2.0 Ken Kennedy (MIT) · DARPA MOCHA / Aarno

Offline / online

Magellan · ACCLAIM · EmitC-MLGO June 2026 PoR

Bring-up / codesign

TritorX · KernelEvolve · Ascend diagnosis · KForge

Kernels / ladder / data

CompileIQ · FlashInfer-Bench ★ · KernelBook ★ · AutoKernel · CuTeGen · Auto ·

Full digest index ~115 entries. Technique map: SURVEY §5.8 T1–T10.

Appendix — publication groups

17 GPU kernels & inference **13** Agentic & RL **13** Source control & review
10 Classic DL compilers **10** Company infra **9** Forums & workshops
8 Surveys & vision **8** Foundation LLMs **6** MLGO & RL gyms
5 HW codesign & bring-up **4** Commercial products **4** Control-plane substrate

How digests are used

Mechanism first; demote generic forge AI; keep both sides — do not average
Update prediction only when Tier A / ★ evidence moves